

ALEX LEWANDOWSKI

Mechatronics Student @ University of Waterloo

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Mechatronics Engineering student at the University of Waterloo (President's Scholarship of Distinction) with four years of competitive robotics and mechanical design experience, including a Top 4 finish out of 20,000+ teams in Autonomous Skills at VEX Worlds. Seeking a Winter 2027 robotics or mechatronics co-op.

EDUCATION

University of Waterloo — B.A.Sc., Mechatronics Engineering Sep 2026 – Jun 2031
President's Scholarship of Distinction

Iroquois Ridge High School — Gifted Program Sep 2022 – Jun 2026
Honour Roll — Top 1% Fryer, Top 2% CCC | 99% admission average (Grade 12), AP Scholar with Honor

TECHNICAL SKILLS

Robotics & Controls: PID, feedforward, odometry, sensor fusion, kinematics, state machines, ROS2

Programming & CAD: Python, C++, Java, Fusion 360

EXPERIENCE

Head Programmer, Designer & Strategist | VEX Robotics Team 5225A, The PiLons Jun 2022 – Jun 2026

- Earned the Design Award at 2025 VEX Worlds, recognizing an organized, professional engineering design process and Engineering Notebook, while ranking 4th of 20,000+ teams worldwide in Autonomous Skills
- Led iterative mechanical design in Fusion 360, redesigning the robot between events to adapt to new strategy
- Built an odometry system fusing gyroscope, encoder wheel, and distance-sensor data to track robot position to ± 0.02 m
- Applied kinematics and inverse kinematics to control robotic arm positioning for precise game-element manipulation
- Designed closed-loop PID and feedforward velocity controllers to drive real-time autonomous motion algorithms
- Developed state machines to coordinate subsystem behavior across autonomous and driver-controlled phases
- Won the Excellence Award at 2025 Ontario Provincials, the top overall team award, given for leading the province across both robot performance and judged criteria
- Won the Think Award at the 2024 and 2026 VEX Worlds, recognizing clean, well-documented autonomous code and a clearly explained programming strategy

Team Captain, Head of Programming | First Lego League Robotics Team Sep 2019 – May 2022

- Programmed autonomous routes (line-following, gyro turns) in Python; planned the route that scored 710 points, 30 above the prior world record
- Led team to Championship at Western Edge (California, 2022) and first-place provincial finishes, including the Engineering & Excellence Award (2020)

PROJECTS

Differential Swerve Drivetrain Sep 2025 – Dec 2025

- Designed each module as a mechanical differential, where summing the two motor speeds drives the wheel and their difference steers it
- Iterated the chassis after load testing exposed C-channel bending, reinforcing it with printed triangle braces and a custom steel-ball thrust bearing

Dexterous Robotic Hand Nov 2025 – Jan 2026

- Designed and 3D printed a five-fingered robotic hand with a rotating wrist and elbow, iterating the finger joint geometry from circular string holes to full-length slits and adding a gear-driven bearing assembly for wrist rotation

World Robot Olympiad Autonomous Robot Oct 2025 – May 2026

- Qualified for the California Invitational Open after Nationals, where the robot was the only one to use pneumatics and custom-designed electronics
- Split the robot's compute between an ESP32 for sensing and localization and a Teensy 4.1 for motor control, integrating a 9-axis IMU, dual color sensors, and a camera module for field-element detection
- Designed and 3D printed a compact chassis in CAD to house the electronics, motors, and sensors within the competition's size limit

LEADERSHIP & ACTIVITIES

Competitive Sailing Skipper — 420 Race Team, Oakville Yacht Squadron; 13+ regattas including Cork International Regatta